Lab 2 – working with Apache Spark (own work)

Laboratory goal:

The aim of the laboratory is to learn how to transform data using the Apache Spark framework. Getting to know different formats for storing data in Apache Spark

Tasks to be completed: → we work on MovieLens data (ml-100k file):

A. Familiarize yourself with the Google Colab environment by going to the project website:

https://colab.research.google.com/

and: Getting Started with PySpark.ipynb (or other) :

https://colab.research.google.com/drive/1G894WS7ltIUTusWWmsCnF_zQhQqZCDOc#scrollTo=1qV6Grv7qIa9

https://spark.apache.org/docs/latest/api/python/

B. In the Google Colab environment, define a new project for Apache Spark preprocessing:

1. Set Up Apache Spark in Google Colab

Google Colab provides a notebook-style interface with built-in support for Python and other frameworks. Here's how you can set up Spark.

PySpark Installation Guide: https://github.com/dbdmg/pyspark-install

1.1 Set Environment Variables: Configure the environment for Spark and Java:
- Check what version of Java is installed?:

```
!java -version
```

1.2 Install Required Packages: Run the following commands in Colab to install Apache Spark and Java:

```
!pip install pyspark
```

1.3 Check pyspark installation:

```
try:
    import pyspark
    print("pyspark is installed.")
except ImportError:
    print("pyspark is not installed.")
```

Exercise:

1. In your browser (Chrome or FireFox), go to the website:

https://grouplens.org/datasets/movielens/

download a sample data set from the website: **ml-100k.zip**

2**.1 Upload Movie Lens dataset files to Google Colab** → The **MovieLens** dataset (**ml10k.zip**) can be downloaded from: https://grouplens.org/datasets/movielens/

(Upload the **u.user**, **u.data**, and **u.item** files to the **Colab directory** → e.g., /content.)

2.2 Load Files into Spark: Load the u.user, u.data, and u.item files as Spark DataFrames.

3. Join Tables and Process Data and next Visualize the Results Graphically:

- Prepare Data for Visualization: After obtaining the sorted DataFrame, convert it to Pandas for easier plotting.

```
# Convert Spark DataFrame to Pandas
sorted_pd_df = sorted_df.toPandas()
```

Then perform the following tasks:

A. - count how many votes were cast (give the number of individual votes cast) for a given rating (for the rating range from 1 to 5)?

B. - count the number of ratings for each movie?

C. - What movie was rated the most – (i.e. the movie with the most votes)?

D. - Which of the movies had the least number of ratings?

E. - Which (what) movie titles have the highest average rating of all ratings? Additionally, check whether these data are reliable based on numerous ratings of a given film

F. - Which (what) movie titles have the highest average rating of all ratings for which the number of ratings is greater than 10?

Homework:

1. What is the oldest movie with the highest average rating (i.e. above 4 stars) and provide the name(s) of the movie(s).

ATTENTION!!!--In the Hive environment, convert the date so that analysis is possible.

2. What is the worst rated movie (with the most ratings below <2)

Analyze the remaining files of the **ml100k** data set, then use the Hive framework to answer the questions:

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3. What genres of films were watched most often by women and which by men?

4. What genres of movies watched by women were rated the highest?

5. Did women (generally) give higher ratings than men or was it the other way around?

6. What genres of films were most often watched by people from specific professional groups (professions)?

7. People from which professional groups gave the highest ratings to which genres of films?

8. Think about it - is there a connection between the ratings themselves, the rated genres of films, my profession and gender?

C. In the Google Colab environment, define a new project for Apache Spark preprocessing:

1. Analyze the data in the file: trucks.csv, geolocation.csv --> and consider:

- what information is stored in the file and how can this data be used for analysis?

- what will each column contain after importing into Apache Spark?

- are the data of the file: trucks.csv matched to the processing using the information stored in the file: geolocation.csv?

- how many rows will be available in each table if You import this data into Apache Spark?

- what theses (hypotheses) can be made when analyzing these data?

- how can the columns (and data) of this table be transformed so that this data can be processed?

2. Make an appropriate transformation of the data structure in the trucks.csv . To do this, write appropriate instructions.

2. Write example queries to geolocation data:

A.) - What is the total number of miles driven by each driver?

(or: how many miles a given driver has travelled since the beginning of trip recording) ?

B.) - What is the average fuel consumption for each truck? - think about whether it is possible to easily calculate the average fuel consumption with such stored data (the solution will be given during classes)?

3. Calculate the risk factor for each driver's road accident as the ratio of the number of accidents to the total number of miles traveled. Then save the data in tabular form in the riskfactor.csv file format.